Cell Host & Microbe

CellPress

Forum

Immune cell states: Critical building blocks of the plant immune system

Tatsuya Nobori[1,*]
[1]The Sainsbury Laboratory, University of East Anglia, Norwich Research Park, Norwich, UK
*Correspondence: tatsuya.nobori@tsl.ac.uk
https://doi.org/10.1016/j.chom.2025.06.016

Plant immunity emerges as cells, normally dedicated to non-immune functions, transition their states through diverse mechanisms to engage in defense roles. This Forum explores the concept of plant immune cell states, their biological significance, and emerging approaches to study them, highlighting the complex cellular basis of plant-microbe interactions.

Multicellularity in the plant immune system

Our understanding of plant immunity has advanced significantly over the past 50 years—from molecular identification and biochemical characterization of resistance genes to reconstructing complex molecular networks.[1] While plants and animals share some similarities in their innate immune systems, including the nature of non-self-recognition and signaling molecules, fundamental cellular differences set them apart. Animals employ specialized and mobile immune cells, such as monocytes, neutrophils, and natural killer cells, which circulate through the body to detect and eliminate pathogens. In stark contrast, plants lack dedicated immune cell types. Individual plant cells, which are generally immobile, must be immunocompetent and coordinate with each other to protect the tissue from unpredictable pathogen attacks. Recent advances in genomics and cell biology approaches have revealed that pathogen-infected plant tissues comprise various kinds of cells with distinct molecular profiles. This emerging paradigm necessitates a re-examination of our gene- (or gene product-) centric view of plant immunity to incorporate a cell-centric perspective. How can plant cells be categorized into meaningful functional units beyond developmental cell types to better comprehend the complexity of the immune system? Understanding immune cell state diversity, function, regulation, and interactions is crucial for unraveling the complex multicellularity of the plant immune system.

What are cell states in plants?

The distinction between cell types and states remains debated, with no universal consensus. Cells continuously change their molecular profiles as they progress through the cell cycle and respond to various signals.[2] While new cell types typically emerge through cell division, boundaries between cell types and states often blur due to continuous cellular transitions. Here, I define a "cell state" as a variation of a "cell type"—a recurring developmental pattern with distinctive molecular and/or cell biological features—influenced by internal and external factors. In plants, where no dedicated immune cell types have been identified, any cell that activates immune responses is regarded as entering an immune state—a functional state of an existing cell type.

Cell state classification cannot rely solely on molecular profiles. Plant cells from different developmental lineages (e.g., mesophyll and epidermis) may display nearly identical molecular signatures during immune activation, overriding their cell type molecular identity. Despite this molecular similarity, it would be more informative to consider these cells as distinct cell states because they originate from different developmental cell types with distinct lineages and spatial locations. These cells can also diverge in their response at later time points, as immune responses are dynamic processes. Thus, a robust cell state definition requires integrating current molecular and positional information with cellular history, such as developmental trajectories and past exposure to environmental stimuli.

Ultimately, no two cells are identical at the molecular and spatiotemporal level. While individual plant cells may be unambiguously defined in some cases (e.g., early embryogenesis), categorizing cells remains essential for studying basic units of multicellular organisms with current experimental and computational tools. Thus, a cell state can be pragmatically defined as a group of cells within a cell type sharing reproducible molecular and positional features and cellular history. This classification is inherently flexible, shaped by both research questions and available technologies.

What shapes plant immune cell states?

Here, I will discuss three factors that influence immune cell states (Figure 1A).

Pre-existing immune potential

Even before the pathogen challenge, plant cells possess varying capacities to mount immune responses—their pre-existing immune potential. This potential might be primarily shaped by developmental cell types, which have specific wiring of molecular pathways, including specific expression of immune receptors, to conduct specific functions. Cell-type-specific plant responses to pathogens have been shown through imaging and cell-type-enriched omics analyses.[3] Cell-type-specific immune potential can also be influenced by abiotic factors (e.g., temperature and nutrient availability) and internal cues (e.g., cell cycle phases). Thus, a plant cell's immune state reflects both its developmental trajectory and its history of environmental inputs. The field has fragmented knowledge about the diversity of pre-existing immune potentials and their underlying mechanisms.

Cell-autonomous and non-cell-autonomous immune responses

Pathogen infection creates two fundamentally different situations in plant cells: cells